SOP: using the NCD PEN-Plus system

Version 1.0 · 2026-11-09 · NCD PEN-Plus clinic (PEN-Plus study)

Purpose. How a subject moves through the PEN-Plus clinic in this system, from the first contact to regular follow-up, and who records what at each step.

Who uses it. Everyone who records subject data at a PEN-Plus clinic: registration officers, nurses, clinicians, laboratory, imaging and pharmacy staff, and data clerks.

Words used. The **PEN-Plus clinic** is the PEN-Plus study: everyone enrolled here is a study **subject**. It has three clinics, one per disease: the **Diabetes Clinic**, the **Cardiac Clinic** and the **SCD Clinic**. Each is both the programme a subject is enrolled in and the clinic they attend; the clinics run on different dates, when their specialists come, and local staff may be trained for one or more of them. A subject can be in more than one clinic. Routine care outside PEN-Plus is recorded in GoTHOMIS. A **PEN-Plus site** is a hospital where the PEN-Plus clinic runs (Kondoa, Karatu); other hospitals (for example BMH, KCMC) are referral or mentoring hospitals. An **appointment** is a booking for a future date. A **visit** (an *encounter* in some menus) is a contact that happened. Everything recorded for a subject belongs to a visit.

1. The flow at a glance

```
New subject   Register → Screening → Investigations → Diagnosis → Prescribe → (Admission) →
Enroll → Follow-up plan
              (all one BASELINE visit; it closes by itself at Enroll)

Enrolled      Appointment day → Attend → FOLLOW-UP visit: Forms → Investigations → Prescribe
→ (Admission) → Close
              Between appointments → UNSCHEDULED visit: Forms → (Investigations) →
(Prescribe) → (Admission) → Close
```

Step	Who	Where in the system
Register	Registration officer, nurse, clinician, data clerk	Subjects → Subject register → **Register subject**
Screening	Nurse, clinician, data clerk	Subject page → **Baseline** → Screening
Request investigations	Nurse (laboratory), clinician (laboratory and imaging)	Baseline → Investigations, or inside a visit
Collect sample, enter result	Laboratory staff	Laboratory → To collect / Awaiting results
ECG, echo, imaging report	Imaging / ECG / echo staff	Radiology & imaging → Requests
Review results	Clinician	Laboratory → Doctor review; Radiology & imaging → Reports to review
Diagnosis	Clinician	Baseline → Diagnosis
Prescribe	Clinician	Baseline → Prescribe, or inside a visit
Dispense	Pharmacist / dispenser	Pharmacy → Dispensing
Enroll	Clinician	Baseline → Enroll
Follow-up schedule (per programme)	Clinician	Baseline → **Follow-up plan** tab
Vital signs and forms	Nurse, clinician, data clerk	Inside the visit → Forms

Step	Who	Where in the system
Admission and discharge	Clinician	Inside the visit → Admission / Discharge
Close the visit	Nurse, clinician, data clerk	Inside the visit → Close visit

The subject page has three buttons at the top: **Baseline**, **Follow-up** and **Unscheduled**. Each shows its steps; a ticked step is done, a greyed step is not needed, and **Next step** links to whatever is still missing.

2. New subject: the baseline visit

Do not book an appointment for a subject seen today. Appointments are only for future dates. Steps 2 to 6 are one **baseline visit**: the system opens it at the first step, links every record to it, and closes it at enrollment.

Vital signs and baseline forms (demographics, history, symptoms and examination, socio-economic status …) are filled during the same visit, under Baseline → **Forms** tab, as soon as the subject has consented. Vital signs are the first form.

Step 1. Register

1. Subjects → Subject register → **Register subject**.
2. Enter the name, sex, date of birth (or age), phone, residence and next of kin. The **Registration facility** lists only the active PEN-Plus sites you work at; with one, it is filled in for you.
3. If the system shows **possible duplicates**, open them. Use the existing subject if it is the same person; register a new one only if it is not, and the reason is recorded.
4. Save. The subject gets an NCD number and their page opens.

Step 2. Screening (once per subject)

1. On the subject page: **Baseline** → **Screening**.
2. Record **consent**. The **Outcome** follows it: "Continues in PEN-Plus" with consent, "Does not proceed at this time" without. Change it only to "Referred elsewhere". - **No consent: stop here.** Nothing else is recorded. The subject continues routine care in GoTHOMIS.
3. Record residence. If the subject already has a known NCD (known diabetic, referral, transfer-in), tick **Known NCD** and the disease(s).
4. **Investigations requested?** - **Yes:** tick under **Screen for** what the tests will look for (diabetes, cardiac, sickle cell, or several). A known disease is ticked for you. Saving opens the investigations page. - **No:** the Investigations step is greyed out and saving goes to **Diagnosis**.
5. The screening form is added once. Afterwards it can only be viewed, edited or deleted (Baseline → **Screening** tab).

Step 3. Investigations

1. Request the tests on the investigations page: laboratory (HbA1c, glucose, Hb, creatinine, sickling test, Hb electrophoresis …) and ECG, echo or imaging.
2. **Laboratory staff** take the sample (Laboratory → **To collect**) and enter the results (Laboratory → **Awaiting results**). Results done in the hospital laboratory are taken from GoTHOMIS and entered here by hand.
3. **Imaging staff** enter the ECG, echo or imaging report (Radiology & imaging → **Requests**).
4. The **clinician** reviews them (Laboratory → **Doctor review**; Radiology & imaging → **Reports to review**).

5. While results are pending, the step shows **Waiting**. If the subject must come back for a test (for example an echo next week), book an appointment for that day and continue from step 4 then.

Step 4. Diagnosis (clinician; once per subject)

1. **Baseline → Diagnosis**.
2. Tick each condition (and sub-type) found, and choose a **status** for each disease. Nothing is pre-selected: - **Confirmed**: the results support it. - **Suspected**: not yet supported, for example a known NCD with no result yet. - **Ruled out**: the disease was looked for and not found.
3. A ticked **Other** condition (for example *Cardiomyopathy – Other*, *Other diabetes*, *Other haemoglobinopathy*) needs a short description in the box under it: say what it is. It is saved as part of the diagnosis.
4. Save. This also gives the screening its result: confirmed counts as positive, ruled out as negative.
5. **No disease found:** press **Not diagnosed** on each screened disease (on the progress bar under Baseline) instead of recording a diagnosis. When every screened disease is not diagnosed, the baseline visit closes as *screening only*, and the subject continues routine care in GoTHOMIS.
6. Afterwards the diagnosis can only be viewed, edited or deleted (Baseline → **Diagnosis** tab). When a suspected diagnosis is later supported by results, edit it to Confirmed.

Step 5. Prescribe (clinician; optional)

1. **Baseline → Prescribe**, if treatment starts today. Add one or more medicines on the same prescription.
2. The pharmacist dispenses it (Pharmacy → **Dispensing**).

Admission (optional). If the subject has to be admitted during the baseline visit (for example in DKA), use **Baseline → Admission** (section 5). While admitted the step shows **Discharge**.

Step 6. Enroll (clinician)

1. **Baseline → Enroll**. The page lists the clinic of every **confirmed** disease (Diabetes Clinic, Cardiac Clinic, SCD Clinic), ticked together.
2. Save. One save enrolls the subject in the clinic(s) and in the PEN-Plus study; the screening consent is the study consent.
3. The baseline visit closes by itself. There is no Close button for it, because it stays open across days while results come back; a line under the Baseline steps says whether it is open or closed. Enrolling is what puts the subject into the indicators, the HbA1c due list and the lists of disease-assigned staff.

Enroll the same day? Yes, if the clinician confirms the diagnosis that day. No, if the subject is suspected and waiting for tests: book the review appointment and enroll then. A suspected diagnosis or a known NCD alone is not enough to enroll.

Step 7. Follow-up schedule (clinician; after enrollment)

1. **Baseline → Follow-up plan** tab (it opens only after enrollment).
2. Tick the **programme clinics** the schedule is for. A subject in one programme has only that one, already ticked. A subject in two or more (for example diabetes and cardiac) can have one schedule for all of them, or one schedule per programme, for example the diabetes clinic monthly and the cardiac clinic every 3 months.
3. The study is PEN-Plus, already chosen. Choose the **interval** (monthly, every 2, 3 or 6 months, or yearly), the start date (the enrollment date by default) and the **number of future visits** (1 to 12; it starts at 1).

4. Save. The system books the appointments; each shows the programme(s) it is for. Subjects who consented to SMS get reminders.
5. Each programme follows **one** active schedule. Ticking a programme on a new schedule moves it from its old one; a schedule left with no programme stops and its future appointments are cancelled. Extra one-off appointments are also booked here (**Book appointment**).

3. Follow-up visit (subject has an appointment)

1. Open the subject page → **Follow-up** → **Attend**. (Or Care → Appointments, or the clinic day page.) - A subject who comes a few days early or late still uses **Attend** on that appointment, so it counts as attended, not missed.
2. The visit opens, linked to the appointment. Its steps:
3. **Forms**: vital signs first, then the forms listed for the visit (Required or Optional). Mark a form **N/A** with a reason if it does not apply. Watch for the HbA1c banner when a test is due.
4. **Investigations**: request the tests for this visit; results follow the same path as at baseline. Every visit makes its own requests: a test requested at an earlier visit (still waiting for its result) is shown for information, and **Already requested** appears only for a test requested earlier in the same visit.
5. **Prescribe**: the medicines for this visit.
6. **Admission**: only if the subject is admitted from this visit (section 5).
7. **Close visit** when the work is done.
8. The appointment changes to **Attended**. The next appointment already exists from the follow-up plan. If the clinician wants a different date, **reschedule** it (a reason is required); do not book a second one.

Subject does not come. Leave the appointment. It becomes **Missed** when attendance for the clinic day is closed, or staff mark it Missed. Follow up by phone or home visit. Missed appointments feed the loss to follow-up (LTFU) count.

One follow-up visit per schedule is expected per month. A subject with a diabetes and a cardiac schedule can attend both clinics in the same month. A second visit for the same schedule, or one with no appointment, raises a data issue (section 7).

4. Unscheduled visit (no appointment)

For example: unwell, ran out of medicine, a complication, referred back between visits.

1. Subject page → **Unscheduled** → **Start unscheduled visit**. It is never linked to an appointment.
2. Same steps as a follow-up visit: **Forms** (vital signs first), **Investigations** and **Prescribe** (both optional here), **Admission** if needed, **Close visit**.
3. It does not attend or replace the next appointment. Keep the next appointment, or reschedule it if the clinician wants the subject back on another date.

If the subject comes **on the day of a booked appointment**, it is not unscheduled: use **Attend** (section 3).

A visit left open on an earlier day does not stop a new one; close it once its work is done.

5. Admission and discharge

1. Inside the visit (baseline, follow-up or unscheduled): **Admission** → ward, date and reason. The subject's Admission tab and Care → Admissions list it.

2. While admitted, the same step shows **Discharge**: record the discharge date, outcome and summary.
3. **Admitted more than once.** A subject can have many admissions, one at a time: discharge the current stay before admitting again. A new admission is recorded from the visit at which the subject comes back (usually an unscheduled visit). If the last stay ended within 30 days, **Readmission of** is filled in with it; clear it if the new admission is not related. Every stay is listed on the subject's Admission tab and under Care → Admissions.
4. A subject who dies in hospital: discharge with outcome **Died** and give the **date and cause of death**. Saving also records the death on the subject: the status changes to deceased, programmes and the study close, and future appointments are cancelled.

6. Outcomes and leaving

All from **More actions** on the subject page, at any time:

- **Transfer to another PEN-Plus site** (for example Kondoa to Karatu): the subject moves and stays in the study. Only active PEN-Plus sites are offered.
- **Refer to another hospital** (for example BMH or KCMC, or any hospital written in): the subject **stays enrolled** and comes back. Record the referral date, hospital and reason; when the subject is back, open the referral (Admission tab → Referrals → **Return / feedback**) and record the date back and the feedback.
- **Transfer out of PEN-Plus**: the subject leaves the PEN-Plus clinic for care elsewhere. Every open clinic closes as *transferred out*, the study enrollment closes, and future appointments and schedules stop. If the subject comes back later, use **Return to care**.
- **Outcome** for one clinic: LTFU, stopped, and later **Return to care**.
- **Record death**.
- **Withdraw / leave the study**: closes the care enrollment here too, and the subject continues routine care in GoTHOMIS. Closing one disease's programme closes the study only if no other programme is active.

7. Checks and corrections

- **Subjects → Stuck in the flow** lists subjects waiting between two steps: screened over 14 days with no decision, positive with no diagnosis, diagnosed but not enrolled, enrolled without a baseline visit, and a programme with no follow-up schedule and no upcoming appointment. Work through it every week.
- **Suspected diagnoses**: if confirming results have not come within 60 days, a data issue says the diagnosis is still suspected; when results arrive, it says "results available: confirm the diagnosis".
- **Correcting**: edit the record. **Removing a duplicate entry**: delete it (a reason is asked). Only a superuser can restore a deleted record.
- **Screening for another disease later**: edit the screening form and tick it. **Screening again** after a negative or inconclusive result: Overview → Screening card → **Re-screen** on that disease, with the reason. The earlier record is kept.

Two follow-up visits in the same month

When a subject has more baseline or follow-up visits in one month than schedules (two visits for the same schedule, or a visit with no appointment next to another), the system assumes the extra ones were unscheduled and raises **"2 follow-up visits in "** under Data quality → Data issues.

1. Open the issue; it lists every visit that month.
2. Keep the real monthly visit of each schedule (the one linked to its appointment).

3. Click **Unscheduled** on each extra visit. It becomes Unscheduled and is unlinked from its appointment; an appointment left with no visit is cancelled ("Visit reclassified as unscheduled during data review").
4. The issue closes by itself when one follow-up visit per schedule is left.

If none of the visits has an appointment, click **Monthly visit** on the planned one; the system creates an attended appointment for it. If the visits are right as recorded, close the issue with **No change needed** and a note. Admissions and screening are not counted in this check.

8. Clinic days (specialist clinics)

When specialists come for a few days or weeks, all active subjects of one programme are booked together, for example the Diabetes Clinic this week, the Cardiac Clinic next week and the SCD Clinic next month. One clinic day is for one clinic.

1. Care → **Clinic days** → **New clinic day**.
2. Choose the **clinic**: Diabetes Clinic, Cardiac Clinic or SCD Clinic. The subjects of that programme are invited.
3. Give the first day and, for a clinic over several days or weeks, the last day and the start time. Choose whether everyone comes on the first day or is spread over the days, and whether existing appointments that month move to the clinic day. Add the specialists' names under **Team**, and the contact phone for subjects.
4. Check the preview, then **Schedule appointments**. The site's staff get an SMS and an email with the clinic, the dates and how many subjects are booked.
5. **Send SMS invitations** to the booked subjects. Subjects who said No to SMS never receive one.
6. **Dates changed?** Edit the clinic day. The booked appointments move to the new dates, staff are told again, and the subject invitations are sent again with **Send SMS invitations**.
7. On the day, subjects are attended from their appointment (**Attend**). After the clinic, **Close attendance** marks those who did not come as missed.
8. **Cancel** a clinic day with a reason: its appointments are cancelled and staff are told. **Notify site staff** sends the staff notice again at any time.

The staff notice goes to active staff assigned to that facility who have a phone number (SMS) or an email address on their profile. It names no subject.

9. Sites (system administrator)

Administration → **Sites** lists every hospital. **Add site** or **Edit** to set:

- **PEN-Plus clinic site**: subjects are registered and seen here. Offered for registration, visits, appointments, clinic days and schedules, to staff assigned to that site.
- **Active**: an inactive site is not offered for new records; its old records keep it.

An active site that is not a PEN-Plus clinic site (a referral or mentoring hospital) is offered only when referring or transferring a subject out of PEN-Plus.

Reference: appointment and visit

	Appointment	Visit
What it is	A plan: "come back on 15 Nov"	A fact: the subject was seen on this date
Created by	The follow-up plan, a clinic day, or staff booking it	Staff, on the day the subject is seen

	Appointment	Visit
Holds	Date, time, visit type, facility	Vital signs, forms, investigations, medicines, admission, notes
Statuses	Scheduled, confirmed, rescheduled, cancelled, attended, missed	Open, in progress, completed, incomplete, cancelled

- An appointment linked to a visit becomes **Attended** automatically. One with no visit becomes **Missed**.
- An appointment with no visit linked (cancelled or never attended) can be deleted; one with a visit is cancelled or marked missed instead.
- Appointments count expected and missed visits, loss to follow-up and SMS reminders. Visits hold the clinical data.

Reports

- **Follow-up & retention** shows, per month: visits due, attended, missed, attendance %, **unscheduled visits**, LTFU, returns and deaths.
- Attendance counts appointments only. Unscheduled visits are shown separately and do not raise or lower attendance.
- Visits migrated from EDCS as UV (unscheduled visit) are imported as **Unscheduled**.

Roles that only read results

PIs and co-investigators, facility and geographic coordinators, statisticians and M&E officers get their own home page instead of the clinic dashboard. It shows reports, indicators, monitoring and subject summaries, view only, with subject IDs and no names. They don't see the subject register, forms or data entry. Statisticians also get data exports and the data dictionary.

Diagnoses imported from EDCS

The old EDCS had a diagnosis form for each disease (diabetes, cardiac, sickle cell) and a "Patient category" form. They were imported into the one Diagnosis record: one diagnosis per diabetes or sickle cell form, one per cardiac condition ticked, and the patient category only where no disease form existed ("category recorded in EDCS"). The old forms are also kept as form records.

EDCS saved every diagnosis as final. Here an imported diagnosis stays **Confirmed** only when a supporting test was imported: echo for cardiac, HbA1c or fasting / random glucose for diabetes, a confirmatory test or genotype (Hb electrophoresis) for sickle cell. Without one it is **Suspected**. The test being done is what counts, not its value, because a treated subject can have normal results. Subjects with a suspected diagnosis stay enrolled; for those still active in care, a data issue asks the doctor to confirm the diagnosis.

Forms from EDCS, and what replaced them

Each EDCS form either carries on as a form here, with the same variable names and answer codes, or its data now lives in a structured record. Every old entry is kept: forms that are no longer filled stay readable with their EDCS entries.

EDCS form	Here
Demographic, Socioeconomic status, Patient history & family history	Forms at baseline, as before

EDCS form	Here
History, symptom & exam	**Symptoms & examination** at every visit; its lab values (fasting / random glucose, HbA1c, Hb, WBC, Plt) are now requested under Investigations
Risk, Diagnosis complications & comorbidities, Hospitalization	Forms at every visit (Hospitalization at follow-up and unscheduled visits)
Treatment plan	**Treatment plan** at every visit, without the medicine rows and insulin doses: those are prescribed under Prescribe
Lab details	Lab values go to Investigations; its assessments (coping, family planning, eye and foot exams, DKA episodes, cardiac surgery, transfusions, transcranial Doppler) are the **Clinical assessments** form at every visit
Main diagnosis – diabetes	The type of diabetes is chosen in the Diagnosis step; presentation (DKA, ketosis…) and hypertension are the **Diabetes presentation** form at baseline
Main diagnosis – cardiac, sickle cell; Patient category	The Diagnosis step (conditions and sub-types, with status); the forms are no longer filled
Results at enrollment	ECG and echo under Investigations; sickle cell tests as lab results
Vitals	Vital signs, the first form of every visit
Hospitalization details	Admission, from inside a visit
Sickle cell family status	**SCD family history** tab (shown for subjects with sickle cell)
Next visit summary	Appointments and the follow-up plan; outcomes (LTFU, transfer, death) under More actions
Screening	The Screening form at baseline

Exporting with the old EDCS names

Exports use the new variable names by default. Choose **Variable names: Old EDCS names** on the Exports page, or **Codes, old EDCS names** in a form's Data menu on the Forms page, to get the EDCS column names instead (for example `dystolic`, `pr`, `clinic_date`, `vid`, `site_id`, `study_id`), so analysis scripts written for the EDCS data keep working. In that mode the subject's NCD number is `ncd_number`, the old EDCS client number is `patient_id` (`id` in the registration table), and lab results take their EDCS `lab_details` names (for example `cre`). A variable EDCS did not have keeps its new name. The data dictionary has an **EDCS table.column** column showing where each variable came from.

Data dictionary versions

Exports → Codebook → **Data dictionary** downloads the versioned Excel data dictionary: study forms, datasets, every variable (label, type, unit, codes, required, in de-identified export, old EDCS name), lab tests, the open decisions (D1–D10, L1–L8) and the change log. Earlier versions, as they were issued, are listed under it. The data team raises the version when forms or variables change.

Change log

Version	Date	Changes
1.0	2026-11-09	First version: the subject flow (registration, screening, investigations, diagnosis, prescribing, admission, enrollment, follow-up schedules per programme), follow-up and unscheduled visits, outcomes, checks and corrections, and the EDCS forms and exports.